

Electromagnetic theory: Activity 1:

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Question #1

$$a) A = \rho \sin \phi \, a_\phi + \rho \cos \phi \, a_\rho - 2z \, a_z \quad \text{--- (i)}$$

$$\rho = \sqrt{x^2 + y^2}$$

$$a_\rho = \cos \phi \, a_x + \sin \phi \, a_y$$

$$a_\phi = -\sin \phi \, a_x + \cos \phi \, a_y$$

$$a_z = a_z$$

Substituting unit vector into eq(i):

$$A = \rho \sin \phi (\cos \phi \, a_x + \sin \phi \, a_y) + \rho \cos \phi (-\sin \phi \, a_x + \cos \phi \, a_y) - 2z \, a_z$$

Simplifying components:-

$$A_x = (\rho \sin \phi \cos \phi - \rho \sin \phi \cos \phi) = 0$$

$$A_y = (\rho \sin^2 \phi + \rho \cos^2 \phi) = \rho = \sqrt{x^2 + y^2}$$

$$A_z = -2z \, a_z = -2z$$

So,

$$A = \sqrt{x^2 + y^2} \, a_y - 2z \, a_z$$

$$b) \vec{B} = 4r \cos \phi \, dr + r \, d\phi$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

$$\cos \phi = \frac{x}{\sqrt{x^2 + y^2}}, \quad \sin \phi = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\cos \theta = \frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad \sin \theta = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

• dr :

$$dr = \sin \theta \cos \phi \, dx + \sin \theta \sin \phi \, dy + \cos \theta \, dz$$

• $d\phi$:

$$d\phi = \cos \theta \cos \phi \, dx + \cos \theta \sin \phi \, dy - \sin \theta \, dz$$

Substitute into vector $\vec{B}$:

$$\vec{B} = 4r \cos \phi (\sin \theta \cos \phi \, dx + \sin \theta \sin \phi \, dy + \cos \theta \, dz) + r (\cos \theta \cos \phi \, dx + \cos \theta \sin \phi \, dy - \sin \theta \, dz)$$

Now:

$$B_x = 4r \sin \theta \cos^2 \phi + r \cos \theta \cos \phi$$

$$B_y = 4r \sin \theta \cos \phi \sin \phi + r \cos \theta \sin \phi$$

$$B_z = 4r \cos \theta \cos \phi - r \sin \theta$$

Substituting coordinate terms:-

$$B_x = 4 \left(\sqrt{x^2 + y^2 + z^2} \right) \left(\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \right) \left(\frac{x}{\sqrt{x^2 + y^2}} \right)^2 + \left(\sqrt{x^2 + y^2 + z^2} \right) \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) \left(\frac{x}{\sqrt{x^2 + y^2}} \right)$$

$$B_x = \frac{4x^2}{\sqrt{x^2 + y^2}} + \frac{xz}{\sqrt{x^2 + y^2}} = \frac{4x^2 + xz}{\sqrt{x^2 + y^2}} = \frac{(4x + z)x}{\sqrt{x^2 + y^2}}$$

$$B_y = 4 \left(\sqrt{x^2 + y^2 + z^2} \right) \left(\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \right) \left(\frac{x}{\sqrt{x^2 + y^2}} \right) \left(\frac{y}{\sqrt{x^2 + y^2}} \right) + \left(\sqrt{x^2 + y^2 + z^2} \right) \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) \left(\frac{y}{\sqrt{x^2 + y^2}} \right)$$

$$= \frac{4xy}{\sqrt{x^2 + y^2}} + \frac{zy}{\sqrt{x^2 + y^2}} = \frac{y(4x + z)}{\sqrt{x^2 + y^2}}$$

$$B_z = 4z \left(\frac{x}{\sqrt{x^2 + y^2}} \right) - \sqrt{x^2 + y^2} = \frac{4xz - (x^2 + y^2)}{\sqrt{x^2 + y^2}}$$

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final answer:-

$$B = \frac{z(4x+z)}{\sqrt{x^2+y^2}} \partial_x + \frac{y(4y+z)}{\sqrt{x^2+y^2}} \partial_y + \frac{4xz-x^2-y^2}{\sqrt{x^2+y^2}} \partial_z$$

Question #2:

given that $B = \rho^2 \sin \phi \partial_\rho + (z-1) \cos \phi \partial_\phi + z^2 \partial_z$ so

find $B \cdot \partial_x$ at $(4, \pi/4, -1)$

so,
calculating dot product of cylindrical unit vector with ∂_x

$$\therefore \partial_\rho \cdot \partial_x = \cos \phi$$

$$\therefore \partial_\phi \cdot \partial_x = -\sin \phi$$

$$B \cdot \partial_x = \rho^2 \sin \phi \cos \phi + (z-1) \cos \phi (-\sin \phi) + z^2 \underbrace{(\partial_z \cdot \partial_x)}_{\text{zero}}$$

$$B \cdot \partial_x = [\rho^2 - (z-1)] \sin \phi \cos \phi \quad \text{--- (i)}$$

Now put value of ρ, ϕ, z in the equation (i)

$$B \cdot \partial_x = 18 \left(\frac{1}{2} \right) = \boxed{9}$$